

Al-Tamayuz Electronics Store

Sales Analytics Portfolio Project

SQL Server • Excel Dashboard • Business Insights



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Executive Summary

End-to-end analysis: from relational data model to business recommendations.



300

Customers

100

Products

5,000

Sales Transactions

5

Categories

Project Objective

The project is based on a database containing **300 customers, 100 products, and 5,000 sales transactions** across multiple product categories such as Tech, Sports, Home & Kitchen, Furniture, and Accessories.

A **comprehensive analysis** was performed to examine sales trends, total revenue, product performance, and customer behavior.

The results are presented in an **interactive Excel dashboard**, featuring:

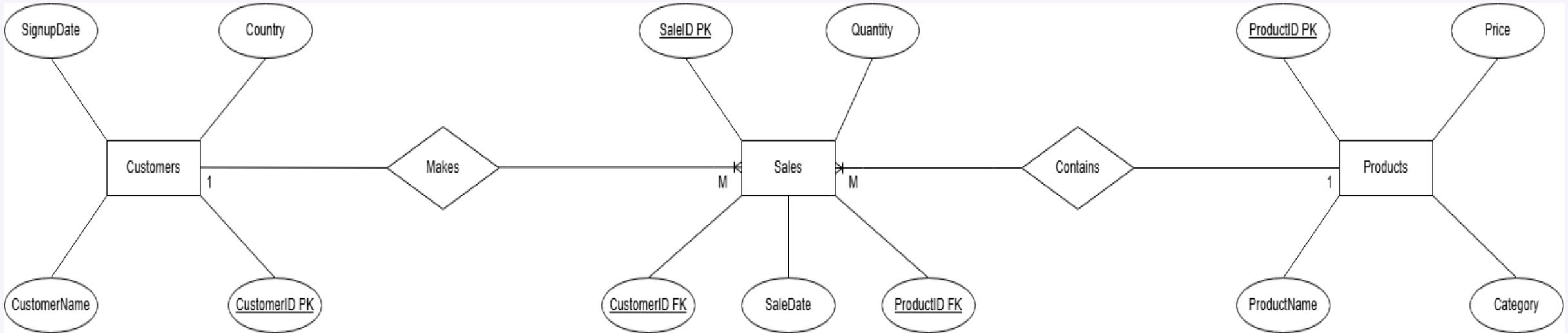
- **Key metrics (KPIs):** Total Revenue, Total Units Sold
- **Visualizations:** Pie Chart showing sales distribution by country and Bar Chart showing revenue by product category

Tools Used

- SQL Server
- Excel
- Power Query
- Pivot Tables & Charts

Data Model & ERD

A normalized relational structure with Sales acting as the bridge table.



Relationship logic

- One customer can have many sales records.
- One product can appear in many sales records.
- Customers and Products have a many-to-many relationship resolved through the Sales bridge table.

Normalization Review

The database structure satisfies 3NF for the scope of this project.



1NF

Atomic values; no repeating groups.

2NF

All non-key attributes depend on the full primary key.

3NF

No transitive dependency between non-key attributes.

Conclusion

- The three-table design is clean for sales analytics because Customers, Products, and Sales each store data about one business concept only.
- No customer or product attributes are repeated in the Sales table, which reduces redundancy and improves query reliability.
- Potential future enhancement: separate Category into a Category lookup table if the business needs category-level ownership, budgets, or hierarchy.

Normalization Review

The database structure satisfies 3NF for the scope of this project.



Customers

<u>Customer_ID</u>	Customer_Name	Country	Signup_Date
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Sales

<u>Sale_ID</u>	<u>Customer_ID</u>	<u>Product_ID</u>	Quantity	Sale_Date
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Products

<u>Product_ID</u>	Product_Name	Category	Price
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This database is normalized up to **Third Normal Form (3NF)**. Each table has a clear primary key, and the Sales table works as a bridge table between Customers and Products. All non-key attributes depend only on their table's primary key, and there are no repeating groups or many-to-many relationships stored directly inside one table.

SQL Analysis Highlights

Core queries used to prepare metrics and segment performance.



Revenue by Category

```
SELECT
    p.Category,
    SUM(s.Quantity * p.Price) AS TotalRevenue
FROM Sales s
JOIN Products p ON s.Product_ID =
p.Product_ID
GROUP BY p.Category
ORDER BY TotalRevenue DESC;
```

High-Value Customers

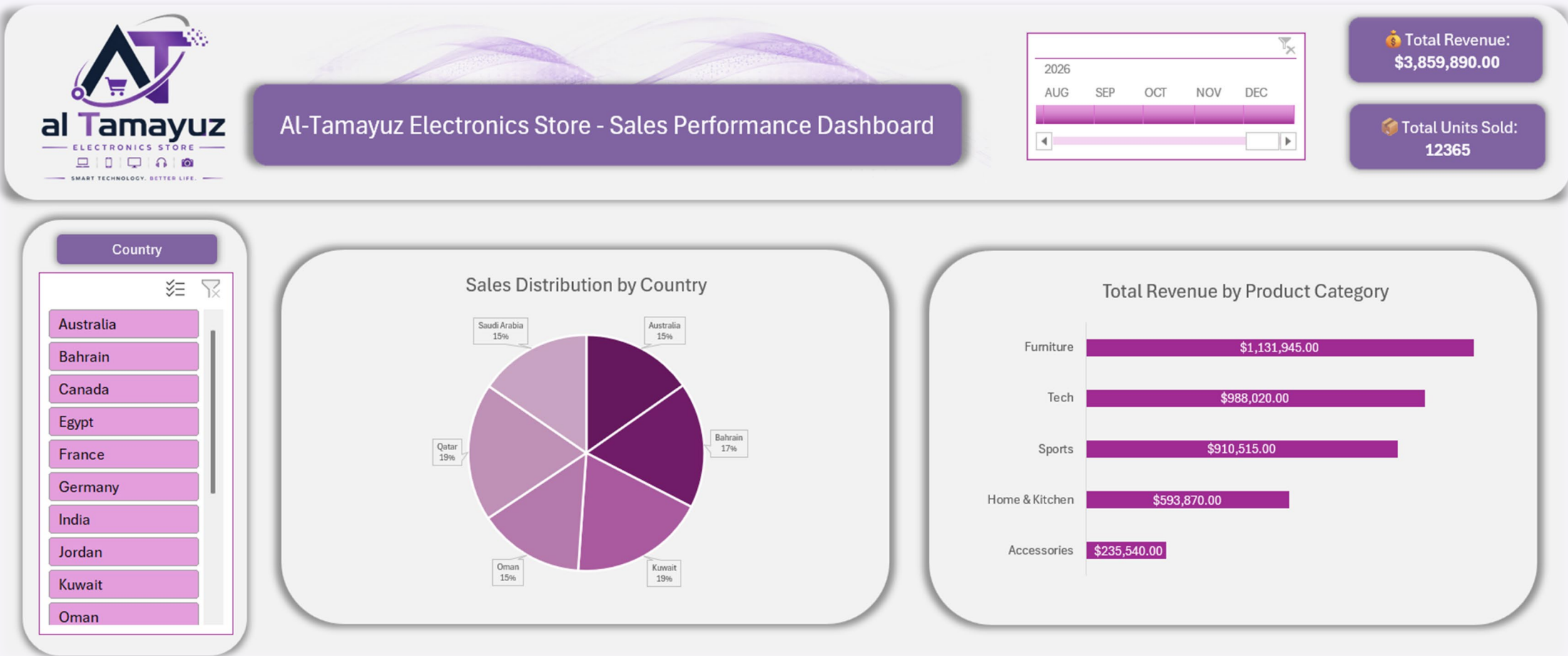
```
WITH CustomerSpend AS (
    SELECT c.Customer_Name,
           SUM(s.Quantity * p.Price) AS
TotalSpend
    FROM Sales s
    JOIN Customers c ON s.Customer_ID =
c.Customer_ID
    JOIN Products p ON s.Product_ID =
p.Product_ID
    GROUP BY c.Customer_Name
)
SELECT *, DENSE_RANK() OVER(ORDER BY
TotalSpend DESC) AS Rank
FROM CustomerSpend;
```

Optimization Plan

```
CREATE INDEX IX_Sales_Customer ON
Sales(Customer_ID);
CREATE INDEX IX_Sales_Product ON
Sales(Product_ID);
CREATE INDEX IX_Sales_Date ON
Sales(Sale_Date);
```

Dashboard Overview

A clean Excel dashboard summarizing sales performance and trends.



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A clean Excel dashboard summarizing sales performance and trends.



\$3.86M

Total Revenue

12,365

Total Units Sold

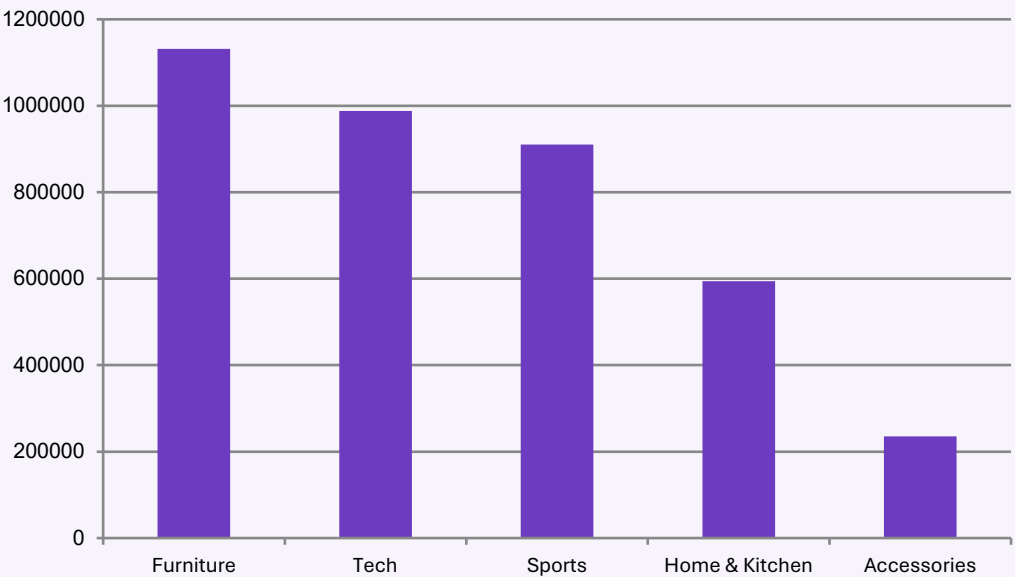
Furniture

Top Category

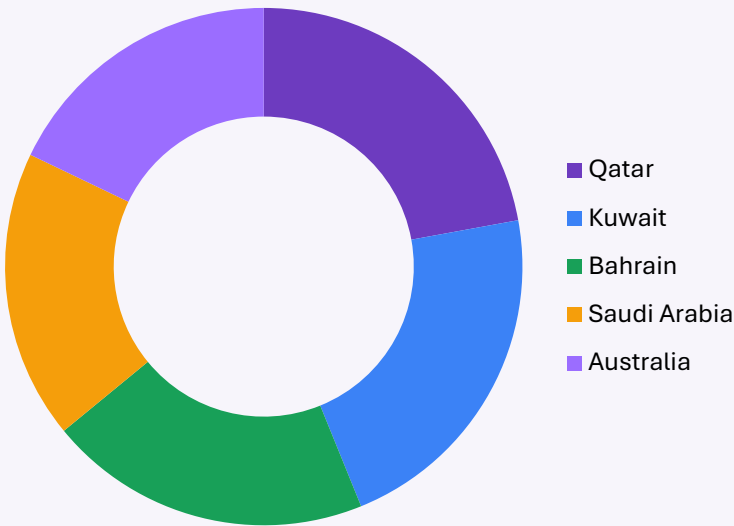
Qatar

Top Country

Revenue by Product Category



Top 5 Countries by Revenue



Key Insights

Sales performance is concentrated in specific categories and countries.



1

Furniture leads revenue

Furniture generated the highest category revenue, followed by Tech and Sports.

2

GCC markets perform strongly

Qatar, Kuwait, Bahrain, and Saudi Arabia appeared among the strongest revenue contributors.

3

Underperforming markets need focus

France, Canada, Germany, and India were among the lowest revenue markets in the dataset.

4

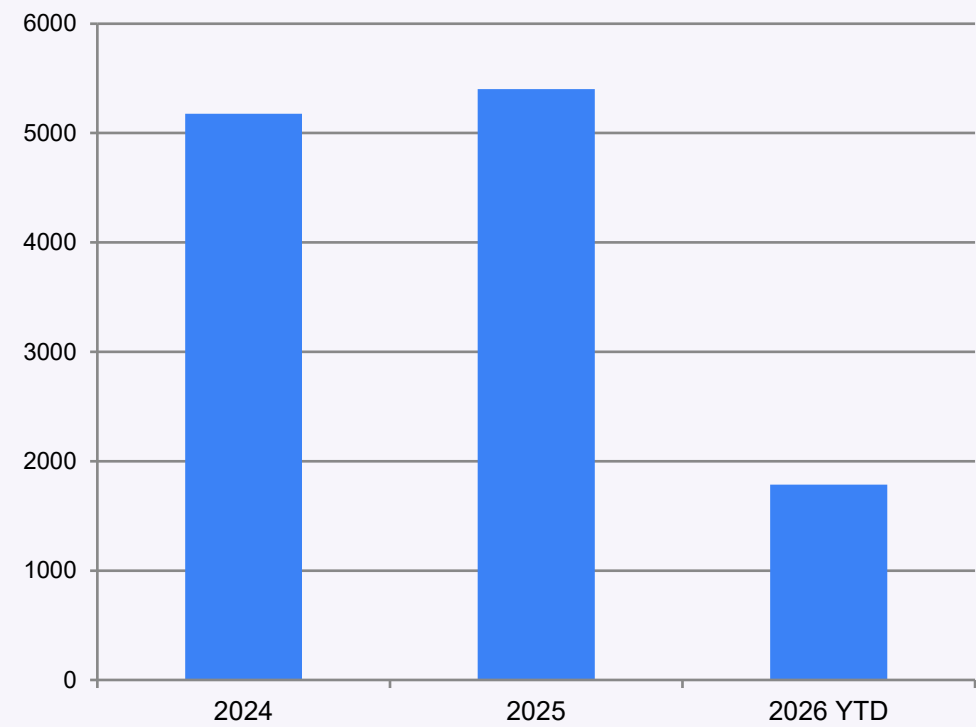
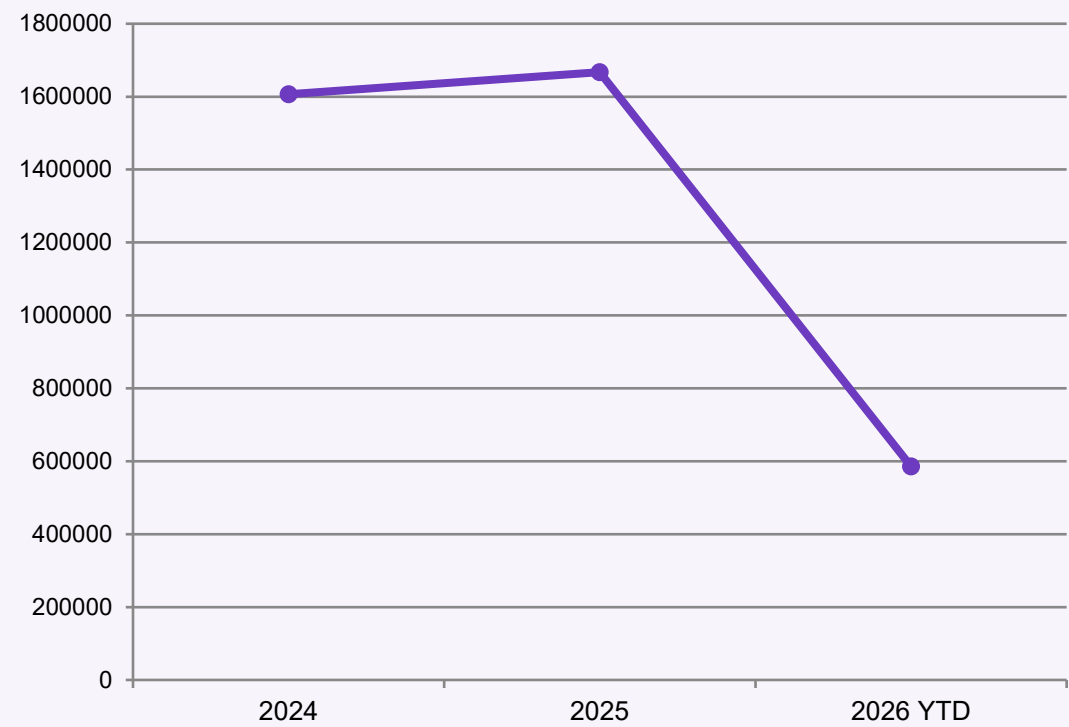
Customer retention matters

Top customers contribute high value, making loyalty and personalized offers important.

Year-over-Year Performance



Revenue improved from 2024 to 2025, while 2026 shows a partial-year decline.



2025 revenue grew by approximately 3.7% versus 2024. 2026 should be treated as YTD/partial-period unless a full year of data is available.

Business Recommendations

Actionable steps based on category, market, and customer performance.



Protect high-performing categories

Maintain momentum in Furniture, Tech, and Sports through targeted offers and stock availability.

Improve weaker categories

Use bundled promotions and campaign testing for Home & Kitchen and Accessories.

Reallocate premium stock

Move premium stock from weaker markets to high-performing countries while maintaining minimum coverage in lower-performing regions.

Build loyalty with top customers

Introduce retention offers, loyalty tiers, and personalized follow-ups for high-value customers.

Scale reporting infrastructure

Index high-usage Sales columns and standardize dashboard refresh steps as the dataset grows.



Thank You

Portfolio Project by Karim Kamel

Open to feedback, collaboration, and data analyst opportunities.

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